Ioannis M. Mandralis

PhD Candidate, Caltech
Graduate Aerospace Laboratories
Center for Autonomous Systems and Technologies
1200 E California Blvd, Pasadena, CA 91106

Education

Ph.D. Aeronautics
Caltech, Sep. 2025
Advisors: Morteza Gharib & Richard M. Murray, GPA: 4.0/4.0, (ranked 1st).

MSc. Mechanical Engineering
ETH Zurich, Sep. 2020
Advisor: Petros Koumoutsakos, GPA: 5.85/6.0.

BSc. Mechanical Engineering
EPFL, Sep. 2018
GPA: 5.78/6.0, 1 year exchange in McGill University, GPA 4.0/4.0.

Research interests

Robotics, multi-modal robotics, model-predictive control, learning, autonomy, fluid dynamics. Special emphasis on aerial robotics with non-trivial morphologies capable of advanced physical interaction. Extensive research on soft robotics and AI for control.

Selected list of Grants and Awards

2022	**Onassis Foundation Scholarship.** Grant awarded for my PhD research in Robotics.
2022	**Rolf D. Buhler Memorial Prize in Aeronautics.** Awarded to student ranked 1st in Aerospace Department.
2021	**Caltech Doctoral Fellowship.** Awarded for PhD studies, tuition & stipend for duration of studies.
2020	**Degree Distinction Prize, ETH Zurich.** Highest degree distinction awarded at ETH Zurich.
2017	**EPFL-McGill mobility.** Fund awarded to top 3 ranked students in EPFL Mech Eng, tuition & stipend for 1y.

Professional Experience

2021-Present — **Ph.D. Researcher,** *Caltech*.
Advisors: **Prof. Morteza Gharib** & **Prof. Richard M. Murray**
Key research project: Learning and control of multi-modal robots capable of dynamic transition between air and ground locomotion in challenging environments.

2018-2021 — **Research Affiliate,** *ETH Zurich/Harvard*.
Advisor: Prof. **Petros Koumoutsakos**
Key research project: Reinforcement learning for control of swimmers in flow simulations.

2019 - 2020 — **Robotics Software Engineer,** *ANYbotics AG, Zurich*.
Developed a novel parameter estimation technique to improve walking performance on the ANYmal robot.

2017-2018 — **Undergraduate Researcher,** *EPFL*.
Advisor: Prof. **John Kolinski**
Optics for two-dimensional photoelasticity, extended for tomography of 3D materials.

2017 — **Summer Intern,** *Nestlé R&D*.
Mechanical Design and Prototyping in the Systems Technology Center, in Orbe Switzerland.

Teaching Experience

Caltech
Head TA, *Optimal Control & Estimation*, Winter 2023. Professor: Richard M. Murray. (link)
Head TA, *Linear Systems Theory*, Fall 2023. Professor: Amir Rahmani.

ETH Zurich
Teaching Assistant, *Multivariable Control*, Spring 2020. Professor: Roy Smith

Phone: +1 (626) 240 9248 | Email: imandralis@caltech.edu | Link: https://mandralis.github.io

EPFL	**Teaching Assistant**, *Thermodynamics*, Spring 2017. Professor: Veronique Michaud **Teaching Assistant**, *Analytical Mechanics*, Fall 2016. Professor: Philippe Mullhaupt

Mentoring Experience

Winter 2025	*Simon Gmur* (MSc visitor: EPFL) MS Thesis: "System Identification for Flying Carpet: a flexible surface for multi-modal dynamic flight."
Summer 2024	*Diego Garcia* (Caltech) Senior Project: "Hardware development for Flying Carpet: A flexible flying surface for multi-modal flight."
Winter 2024	*Quentin Delfosse* (MSc visitor: EPFL) MS Thesis: "Reinforcement-learning based navigation for a multi-modal robot."
Fall 2024	*Gabriel Margaria* (MSc visitor: EPFL) MS Thesis: "Adaptive control of a multi-modal robot using deep reinforcement learning."
Fall 2024	*Vincent Gherold* (MSc visitor: EPFL) MS Thesis: "Self-supervised cost of transport estimation for multi-modal path planning." **Award:** Best Master Thesis Dissertation in Robotics at EPFL (link).
Fall 2023	*Severin Schumacher* (MSc visitor: TUM) MS Thesis: "Experimental analysis of the transition and landing phase of a flyable rover."
Summer 2023	*Sofia Wan* (Caltech) & *Steven Lei* (Caltech) SURF project: "Hardware development for a flexible flying surface." **Award:** Finalist for best Summer Undergraduate Research Fellowship (SURF) project.

Selected Talks/Workshops

2025	*Quadrotor Morpho-Transition: Learning vs Model-Based Control Strategies*. Oral Presentation, IROS, Hangzhou, China.
2025	*Ground-Aerial Multi-Modal Robotics for Automated Delivery*. Invited Talk, HiPeR Lab Berkeley CA.
2024	*Multi-Modal Robotics Booth in collaboration with Technology Innovation Institute (TII)*. IROS, Abu Dhabi.
2023	*Minimum Time Trajectory Generation for Bounding Flight*. ECC, Bucharest, Romania.
2023	*Thrust recapture for morphing aerial vehicles with out of plane thrusters*. APS DFD, Washington DC.
2021	*Investigation of surface pressure fluctuations in airfoils: towards gust mitigation*, APS DFD, Phoenix AZ.

Press

2025	Mid-Air Transformation Helps Flying, Rolling Robot to Transition Smoothly
2024	Sky News Arabia coverage of ATMO at IROS 2024
2021	Engineers Teach AI to Navigate Ocean with Minimal Energy

Service

Peer Review	IEEE Robotics and Automation Letters, Nature Scientific Reports, ICRA, IROS, and ECC conferences.
Outreach	In charge of more than 15 Caltech CAST tours for investors, media, student groups, and general PR.

References

Morteza Gharib
Hans W. Liepmann Professor of Aeronautics and Medical Engineering.
Director of GALCIT and CAST.
Phone: +1 (626) 376 1100, Email: mgharib@caltech.edu, Website.

Richard M. Murray
Thomas E. and Doris Everhart Professor of Control and Dynamical Systems and Bioengineering.
Phone: +1 (626) 487 2418, Email: rmurray@caltech.edu, Website.

John O. Dabiri
Centennial Professor of Aeronautics and Mechanical Engineering.
Phone: +1 (626) 395 4450, Email: jodabiri@caltech.edu, Website.

Publications

Publications listed in chronological order.

[1] **I. Mandralis**, R. Nemovi, A. Ramezani, R. M. Murray, M. Gharib, "ATMO : An Aerially Transforming Morphobot for Dynamic Ground-Aerial Transition", Nature Communications Engineering, April 2025. https://www.nature.com/articles/s44172-025-00413-6.

[2] **I. Mandralis**, R. M. Murray, M. Gharib. "Quadrotor Morpho-Transition: Learning vs Model-Based Control Strategies", Accepted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. https://arxiv.org/pdf/2506.14039.

[3] **I. Mandralis**, S. Schumacher, M. Gharib, "A Method for Thrust Recovery in Aerial Robots", (submitted) June 2025. Preprint available upon request.

[4] V. Gherold, **I. Mandralis**, E. Sihite, A. Salagame, A. Ramezani and M. Gharib, "Self-Supervised Cost of Transport Estimation for Multimodal Path Planning," in *IEEE Robotics and Automation Letters*, vol. 10, no. 7, pp. 6872-6879, July 2025. https://ieeexplore.ieee.org/abstract/document/11012130.

[5] A Salagame, S Pandya, **I. Mandralis**, E Sihite, A Ramezani, M Gharib, NMPC-based Unified Posture Manipulation and Thrust Vectoring for Fault Recovery, (submitted) June 2025, https://arxiv.org/pdf/2503.18307.

[6] AA. Stefan-Zavala, I. Scherl, **I. Mandralis**, SL. Brunton, M. Gharib, "Data-Driven Modeling for On-Demand Flow Prescription in Fan-Array Wind Tunnels", (submitted) February 2025, https://arxiv.org/pdf/2412.12309.

[7] Bibek Gupta, Aniket Dhole, Adarsh Salagame, Xuejian Niu, Yizhe Xu, Kaushik Venkatesh, Paul Ghanem, **Ioannis Mandralis**, Eric Sihite, Alireza Ramezani, "Bounding Flight Control of Dynamic Morphing Wings," 2024 IEEE International Conference on Advanced Intelligent Mechatronics (AIM), Boston, MA, USA, 2024, pp. 100-105. https://ieeexplore.ieee.org/abstract/document/10637050.

[8] Adarsh Salagame, Neha Bhattachan, Andre Caetano, Ian McCarthy, Henry Noyes, Brandon Petersen, Alexander Qiu, Matthew Schroeter, Nolan Smithwick, Konrad Sroka, Jason Widjaja, Yash Bohra, Kaushik Venkatesh, Kruthika Gangaraju, Paul Ghanem, **Ioannis Mandralis**, Eric Sihite, Arash Kalantari, Alireza Ramezani, "How Strong a Kick Should be to Topple Northeastern's Tumbling Robot," 2024 IEEE International Conference on Advanced Intelligent Mechatronics (AIM), Boston, MA, USA, 2024, pp. 76-81. https://ieeexplore.ieee.org/abstract/document/10637166.

[9] **I. Mandralis**, E. Sihite, A. Ramezani and M. Gharib, "Minimum Time Trajectory Generation for Bounding Flight: Combining Posture Control and Thrust Vectoring,"
2023 European Control Conference (ECC), Bucharest, Romania, 2023.
https://ieeexplore.ieee.org/abstract/document/10178360.

[10] Eric Sihite, Filip Slezak, **Ioannis Mandralis**, Adarsh Salagame, Milad Ramezani, Arash Kalantari, Alireza Ramezani, Morteza Gharib, "Demonstrating Autonomous 3D Path Planning on a Novel Scalable UGV-UAV Morphing Robot," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 3064-3069. https://ieeexplore.ieee.org/abstract/document/10342189.

[11] Aniket Dhole, Bibek Gupta, Adarsh Salagame, Xuejian Niu, Yizhe Xu, Kaushik Venkatesh, Paul Ghanem, **Ioannis Mandralis**, Eric Sihite, Alireza Ramezani, "Hovering Control of Flapping Wings in Tandem with Multi-Rotors," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 6639-6644.

https://ieeexplore.ieee.org/abstract/document/10341523.

[12] P. Gunnarson, **I. Mandralis**, G. Novati, P. Koumoutsakos & JO. Dabiri, "Learning efficient navigation in vortical flow fields." Nature Communications 12,7143, (2021). https://www.nature.com/articles/s41467-021-27015-y.

[13] **I. Mandralis**, P. Weber, G. Novati, P. Koumoutsakos, "Learning swimming escape patterns under energy constraints." Phys. Rev. Fluids 6, 093101, (2021). https://journals.aps.org/prfluids/abstract/10.1103/PhysRevFluids.6.093101.

Patents

[1] A. Rosakis, **I. Mandralis**, M. Gharib, "Ultrasound Shape Sensing" patent pending approval of US Patent Office (submitted Jan 2025).

[2] **I. Mandralis**, S. Schumacher, M. Gharib, "Thrust Recapture for Morphing Aerial Vehicles with Out-of-Plane Thrusters," patent disclosure obtained (Sep 2024).